Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Distributed Systems · Java · Fault-Tolerant Services · Backend at Scale · Python · AWS · Observability · AI-Assisted Engineering

venkatalakshmiparimalap@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | AgentProbe (open-source) | Virginia · Open to Cupertino / Remote

PROFESSIONAL SUMMARY

I build **distributed, fault-tolerant backend services at scale — Java + Python, async/multi-threaded systems, observability** — with 9+ years including 7+ years shipping production systems in regulated domains. At Entrata (100K+ MAU multi-tenant SaaS), built **Java/Spring Boot** and Python services processing 1.6M federal compliance records, **RabbitMQ** async/multi-threaded pipelines with idempotency guarantees, and **relational + NoSQL database** systems (PostgreSQL 7 yrs, Redis, Oracle) with 200+ query optimizations. At Intuit, own production services on **AWS** with full observability (OpenTelemetry, Prometheus, Grafana) and integrate Anthropic Claude for **AI-assisted engineering** — the exact workflow Apple lists as a preferred qualification. Open-source: **AgentProbe**, an async multi-provider backend system with observability and **monitoring**; **Leasel**, a distributed agentic pipeline with **test automation** CI (93.6% precision, 1,240 cases). Messaging platforms: **RabbitMQ** async patterns, **Kafka** (familiar), Celery. **Unix/Linux** environments throughout. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Distributed Systems | Fault-Tolerant Services | Java & Spring Boot | Multi-Threaded & Async | RabbitMQ / Kafka Messaging | NoSQL + Relational Databases | Observability & Monitoring | Backend Services at Scale | Caching Strategies (Redis) | AI-Assisted Engineering | Test Automation & CI/CD | Mentoring & Code Review

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — Production Services · AWS · Observability · AI-Assisted Engineering · Python

- Own backend services on **AWS** (EC2, RDS, S3, Lambda, API Gateway) for financial data processing; full **observability** stack: OpenTelemetry instrumentation, Prometheus metrics, Grafana dashboards — production visibility from deploy to alert.
- Build ML pipelines and **Python**/FastAPI services with validation, error handling, and asynchronous processing; containerize with Docker and ship via GitHub Actions CI/CD with automated **test automation** suites.
- Leverage **AI-assisted engineering** daily: integrate Anthropic Claude for financial data summaries and natural-language queries, with structured-output schemas and deterministic validators for downstream service reliability.
- **Monitoring**: LLM-as-judge eval CI caught 3 prompt regressions before release; zero hallucination incidents post-deploy; input validation, sensitive-data handling, and fallback paths on model failure fully operationalized.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, multi-tenant platform

Senior Software Engineer — Java · Spring Boot · Python · Distributed Systems · PostgreSQL · RabbitMQ · AWS

- Built and shipped 20+ production features end-to-end in **Java/Spring Boot**, Python, PHP, and PostgreSQL across 120+ Agile sprints on a **distributed** multi-tenant SaaS platform serving 100K+ monthly active users; reviewed 300+ PRs; mentored junior engineers on architectural patterns, code quality, and testing strategies.
- Designed **fault-tolerant RabbitMQ**-based **asynchronous** processing system for bulk rent updates across thousands of tenant records: **multi-threaded** consumer workers, dead-letter queues, exponential backoff retry, idempotency guarantees to prevent double-processing — messaging platform patterns directly applicable to partner onboarding and pricing pipelines.
- Owned HUD federal compliance ETL: 180K+ regulatory records per annual cycle with validation, rollback checks, and audit trails; processed 1.6M HUD/Tax Credit records in large-scale data import workflows; income fraud verification in affordable housing — business systems requiring financial accuracy and **fault-tolerant** processing at Apple's regulated-domain scale.
- Optimized 200+ **relational database** (PostgreSQL) queries via execution plan analysis and indexing (30% performance improvement); **caching** layer with **Redis** for high-volume async workflows; **AWS S3** PHI migration with Libsodium encryption; Docker microservices redesign (50% scalability improvement); Python/**Unix**/Bash automation eliminating 80% of manual operational work.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Java · Python · PHP · PostgreSQL · AWS · Unix/Linux

- Built property management modules and compliance data pipelines in **Java**, Python, PHP, and PostgreSQL processing 1.6M HUD/Tax Credit records; led **AWS** S3 cloud migration from legacy server storage on **Unix/Linux** infrastructure; created 56 state-specific housing certification documents requiring financial accuracy at scale.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

ML Research · Fine-Tuning · PyTorch · HuggingFace · Statistical Modeling

- Fine-tuned classification model (HuggingFace Transformers + PyTorch): full training loop (data prep → fine-tuning → eval → deploy on Quart), presented to GMU faculty panel. Built predictive models ($R^2 = 0.977$); applied Isolation Forest anomaly detection. Mentored 600+ students.

PORTFOLIO PROJECTS

AgentProbe — Async Multi-Provider Backend System with Observability & Monitoring

Distributed Backend · Async Systems · Observability

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Pinecone, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- **Asynchronous multi-provider backend**: 15-endpoint FastAPI service routing requests across 5 LLM providers with an LLM Router for intent classification and dynamic provider selection based on latency, cost, and capability — **fault-tolerant** with fallback paths and **monitoring** hooks at every provider boundary.
- **Observability** layer: 5-metric evaluation framework (factual accuracy, hallucination rate, retrieval quality, completeness, compliance) with CI-integrated regression detection; **test automation** with golden datasets blocks deploys on quality drops; evaluated Leasel across 1,240 test cases; **NoSQL** vector storage (ChromaDB, Pinecone).

Leasel — Distributed Agentic Pipeline for Compliance Automation

Distributed Pipeline · Fault-Tolerant · Test Automation CI

2026 · Python, LangGraph (8-agent), FastAPI, hybrid RAG (TF-IDF + semantic reranking), SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Distributed agentic pipeline** with 8-agent LangGraph workflow: triage → retrieval → parallel Config/History/StructuredData agents → reasoning → deterministic validator → confidence-gated HITL escalation; **fault-tolerant** design with explicit error states and fallback routing at each agent boundary.
- **Test automation** CI harness: 93.6% precision, 87.9% F1 across 1,240 test cases; LLM-as-judge scoring + regression detection integrated into GitHub Actions; hybrid RAG with TF-IDF + semantic reranking; **relational database** (SQLite) + **NoSQL** vector storage.

AI on FHIR — Clinical NLP Backend Service with HIPAA Guardrails

Backend Services · API Integration · HIPAA Compliance

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP pipeline translating English/Spanish clinical questions into FHIR API calls via LLM tool-calling and embeddings; HIPAA-compliant RBAC for data isolation; ~60% reduction in manual clinical lookup time; containerized with Docker on **Unix/Linux**; React + TypeScript frontend; REST API backend.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Languages & Backend: Java, Spring Boot, Python (9+ yrs), FastAPI, Flask, Django, TypeScript, JavaScript, Node.js, PHP, SQL, PL/SQL, Bash, Kotlin, Scala, GraphQL, C#, .NET

Distributed Systems & Messaging: Distributed Systems, Fault-Tolerant Services, Multi-Threaded Programming, Asynchronous Processing, RabbitMQ, Kafka, Celery, Microservices, REST APIs, Partner Onboarding Pipelines, Event-Driven Architecture

Databases & Caching: PostgreSQL (7+ yrs), MySQL, Oracle, Redis (caching + async), MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization, Stored Procedures, NoSQL, Relational Databases

Cloud, DevOps & Observability: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, Unix/Linux, Test Automation

AI-Assisted Engineering & AI/ML: AI-Assisted Engineering (Claude Code, Anthropic Claude), LLM APIs, RAG Pipelines, LangGraph, LangChain, LLM-as-Judge Evaluation, Prompt Engineering, Guardrails, scikit-learn, XGBoost, PyTorch, HuggingFace, PySpark, ChromaDB, Pinecone, Weaviate, Elasticsearch